

Research report after round 1. The run is still in progress: this is the report text as it stands after this round, text only, with no figures. The results and the writing are not final.

Framing

This run looks for metrics that say whether a first-order logic (FOL) formula is a faithful translation of a natural-language sentence, when no gold formula is available. The setting comes from the user. They build NL-to-FOL pipelines with large language models (LLMs), and their evaluation stalls on two points. First, gold FOL is scarce and not unique: different predicate names, different decompositions of one concept, and logically equivalent rewrites can all be correct. Second, their own pilot metrics measure only stability and composability, not meaning. Those metrics were joint loadability, arity consistency, undefined predicates, and predicate-set overlap across reruns. The target is fixed in advance. A metric has to predict faithfulness to the text on real system outputs, and ideally name the error type. Detecting synthetic perturbations, parse validity and run-to-run stability do not count as substitutes. Every metric is compared with parse rate, round-trip similarity, LLM-as-a-judge, sampling self-consistency and the user's structural metrics. Results are reported by sentence complexity, because long, heavily conditioned sentences matter most to the user.

The public benchmarks that supply gold are themselves unreliable. FOLIO [1], MALLS [2] and ProverQA [3] carry sentence-level FOL annotations. A 2026 relabelling study reports that about 42.5% of FOLIO-validation and 42% of MALLS-test entries contain incorrect formalizations, and that 17.8% and 51% of their sentences are ambiguous [4]. The gold-free methods already in use compare a whole formula, or its back-translation, with the text through an LLM. Examples are round-trip verification [5], clause-conformance scoring [6] and LLM-as-a-judge [7]. A trained reference-free verifier, GenV, reaches 0.961 AUROC on reference-equivalence verification [8]; its task is to predict equivalence to a designated reference, not faithfulness to the text. The closest gold-based ancestor of the run's main idea is the polarity evaluation of SyGNS [9], which compares per-word monotonicity marks of a predicted formula with those of the gold. A second relative matches NL-derived probes against the formal consequences of a statement [10]. The systems whose outputs are scored here are the Logic-LM [11] and LINC-style [12] pipelines: an LLM translates, and a prover or solver such as Z3 [13] reasons.

The run also leans on four other lines of work. First, semantic evaluation of logical forms. Test-suite accuracy for text-to-SQL [21] and its formal-verification critique [35] compare denotations with a gold query, and a perturbation study of FOL closeness metrics [20] measures how gold-based scores react to typed edits. The run reuses typed perturbations only as a diagnostic instrument. Second, distinguishing inputs judged by an LLM: this idea is established for code and SQL [24, 25] and for formal specifications [26], and it is the mechanism behind the truth-value-judgment alternate. Third, agreement among samples. Self-consistency [30], symbolic-equivalence consistency in autoformalization [33], SAT-based selection among compositional translations [31] and grammar-based uncertainty measures [32] all score agreement within a generator's own distribution rather than agreement with the text. Entailment-preserving FOL metrics [34] are reference-free but need multi-premise NLI labels. Fourth, estimation without a gold standard. Latent-class models of rater error [22, 23] come from epidemiology, where they are known to be biased when the tests' errors are conditionally dependent [36]; gold-free benchmarking of LLM judgments has also been proposed [37]. Sentence embeddings for round-trip similarity use Sentence-BERT-style encoders [28] with MiniLM [29].

The hypothesis the run started from is that a formula is faithful only if it shares the sentence's *monotonicity signature*. We use the hypothesis's own definition throughout. For every concept the sentence mentions, the signature records whether the sentence stays true when the concept is generalized (upward), specialized (downward), both (vacuous) or neither. It also records which concept fills each argument slot of each relation (*role anchoring*). On the formula side a solver decides these labels exactly over small finite domains. On the sentence side a formula-blind *substitution probe* asks an LLM whether the sentence entails copies of itself with one concept specialized. The probe was validated independently on the monotonicity NLI sets MED [14] and HELP [15], and the zero-LLM variant uses rule-based polarity marking [16, 17]. The idea borrows from generalized-quantifier theory [18] and from vacuity detection in model checking [19]. Three alternates were carried from the start:

- *truth-value-judgment worlds*: a solver builds small worlds that separate a candidate from its mutants, and an LLM judges only whether the sentence is true in each world;
- *instance-consequence NLI*: a small NLI model checks prover-derived ground consequences against the sentence;
- *latent-class estimation*: the outputs of several systems are treated as imperfect diagnostic tests.

The run allows two iterations. This document records them in order.

Iteration 1

1.1 Where the iteration started: the hypothesis review

Before the invention loop began, a reviewer scored the hypothesis 5/10 (soundness 2, presentation 3, contribution 3, fidelity 4). The blocking flag was set only because no experiment had run yet. The reviewer ran the hypothesis's own Z3 prototype and found four common error classes to which the plain signature is blind:

- conjunction/disjunction confusion in the restrictor;
- conjunction/disjunction confusion in the scope;
- consequents swapped under an exception;
- swapped constants.

Exception concepts also collapsed to the non-monotone label. Since these errors sit in exactly the long, exception-laden sentences the complexity prediction is about, the reviewer argued that per-concept coordinates lose information where the run hoped they would gain it.

The reviewer made four further objections:

1. Invariance "by construction" holds for the solver core only. A lexical aligner reintroduces vocabulary sensitivity: a compound predicate such as `UntrainedDog` aligned to "trained" raises a false polarity alarm.
2. The planned granularity-aware label reused name alignment, which couples the labels to the metric.
3. SyGNS [9], GenV [8] and the probe-matching work [10] were uncited.
4. Nothing in the plan measured what fraction of real errors change the signature at all.

The reviewer asked for three additions: an oracle-ceiling condition that uses the gold formula's

signature in place of the text probe; a decomposed-judge ablation, which asks an LLM the same per-concept questions about the formula; and label construction independent of the metric's aligner. The reviewer also verified a repair, the *relativized signature*: the monotonicity of predicate P when P may change only inside, or only outside, another predicate Q. The relativized signature separated all three failing pairs with 6 to 8 differing coordinates, and it gave 0 differences on a contrapositive rewrite.

1.2 Strategy: a pre-registered wide screen plus a held-out dataset

With four live candidate metrics and two iterations, the strategy step chose a wide screen over depth. It had two reasons. The candidates see different errors, so real outputs had to decide between them. The signature is local and depends on alignment. The world-based checks are global and alignment-free but depend on verbalizing predicate names. The latent-class model ignores the text entirely. If real errors were mostly scope errors, the world-based checks should win; if they were mostly polarity or condition errors on long sentences, the signature should win. Second, a held-out labelled set that was not built in iteration 1 could not be built later, so one of the three artifact slots went to a dataset.

Iteration 1 therefore ran three artifacts in parallel. Screen Arm A tested the signature variants. Screen Arm B tested the three alternates. A dataset artifact built a held-out confirmation set for iteration 2. Both screen arms rebuilt one frozen screen set from public sources with an identical deterministic recipe:

- the Logic-LM released translations of FOLIO-dev by gpt-3.5-turbo, gpt-4 and text-davinci-003 [11];
- gold from FOLIO validation [1], with the 2606.02837 corrections [4] where retrievable;
- the first 300 matched sentences in sha1 order;
- labels from blind Z3 equivalence up to an arity-preserving bijection of predicates and constants, searched without lexical information.

Typed mutants and solver-verified meaning-preserving rewrites of each gold served as gates and mechanism diagnostics, never as the headline.

The selection rule was fixed before any score existed and copied verbatim into both arms. The primary measure was the item-level AUROC on the real candidates against the bijection-equivalence labels, with a $1000 \times$ bootstrap clustered by sentence. Table 1 lists the four gates.

Table 1. The pre-registered survival gates, fixed before any scores existed.

Gate | Condition

G1 coverage | at least 70% of real candidates scored; unscored items get 0.5 and stay in the AUROC

G2 false alarms | at most 10% on solver-verified meaning-preserving rewrites

G3 increment | ΔAUROC of $\text{logistic}(\text{cheap judge} + \text{parse_ok} + \text{metric})$ over $\text{logistic}(\text{cheap judge} + \text{parse_ok})$, clustered-bootstrap 95% lower bound > 0

G4 AUROC floor | AUROC on real candidates ≥ 0.65

Survivors were to be ranked by AUROC on the top complexity tercile, and a winner declared only with a lead of at least 0.03 there. If nothing survived, the candidate with the largest increment point estimate would advance, provided that estimate was at least 0.03. In iteration 2 the rule is to be re-applied with the dataset's audited screen labels, and the audited ranking governs if the two disagree.

The strategy recorded where it broke the field's norms, and why:

1. The screen used unadjudicated equivalence labels on partly uncorrected gold. This is cheap and identical for every candidate. Its known bias, counting correct-but-gold-inequivalent candidates as wrong, works against the granularity-invariant signature.
2. The screen's systems date from 2023. The held-out set was meant to shift both the system era and the corpus.
3. The screen is powered for ΔAUROC of about 0.06. That is enough to choose a survivor, not to claim a finding.
4. Latent-class estimation is thin with three systems.

1.3 Screen Arm A: the monotonicity signature

What was built

The formula side parses FOLIO syntax and grounds each formula propositionally. A single SAT query covers every domain size from 1 to N at once, using an active-element encoding. With $N = 3$, the run computed 2,645 signatures in 19 s. For each predicate the solver decides upward, downward, vacuous or non-monotone, and it decides slot anchors. The relativized coordinates of the review fix are also computed.

Three candidate metrics share this formula side and differ in the text side:

- **LLM-probe signature** (row label A1): text labels come from the substitution probe, gemini-2.5-flash with a 1,024-token thinking budget. The probe answers batched upward and downward entailment questions per sentence and never sees a formula.
- **Relativized signature** (A2): the LLM-probe signature plus relativized coordinates, the review's fix.
- **Rule-marker signature** (A3): text polarity from a compact rule-based monotonicity marker over a spaCy parse, with zero LLM calls. Udep2Mono [17] could not be installed within the time box.

Concepts are aligned to predicates by lemma and camelCase overlap. The aligner has rules for compound heads and lexical negation, with a MiniLM [29] embedding fallback. The score is one minus the weighted fraction of mismatched coordinates. A fixed decision list maps the mismatch pattern to an error type.

Two controls were added that were not in the plan, because a lexical aligner can carry signal of its own. The *alignment-only control* (A0) is the same score with the polarity and anchor coordinates switched off. The *concept-coverage control* (Ccov) is the fraction of sentence concepts that align to any predicate. The baselines were:

- the whole-formula LLM judge (B1): gemini-2.5-flash, thinking off, temperature 0, with a fixed prompt asking for the probability that the FOL is faithful;
- parse rate (B2) and its satisfiable variant;
- round-trip (B3): FOL→NL by gemini-2.5-flash, scored by MiniLM cosine or by bidirectional DeBERTa-v3-large NLI [27];
- the decomposed judge (B4): the same LLM answers the same per-concept upward/downward questions about the formula;

- the gold-signature oracle (B5): the text signature is replaced by the gold formula's solver signature, giving the ceiling if the text side were perfect;
- probe accuracy on MED and HELP (B6);
- the user's pilot structural metrics (B7);
- self-consistency (B8), added during execution: Logic-LM re-translated each story's premises independently for every example of that story, so the agreement of a candidate with the same system's other translations of the same sentence costs nothing.

Screen set as built by Arm A

The screen contains 300 sentences and 835 real candidates: gpt-3.5-turbo 265, gpt-4 298, text-davinci-003 272. Of these, 811 parse. Ten gold formulas were excluded as unparseable; all used the character ' inside predicate names. The labels are complete: 389 equivalent and 446 non-equivalent, with 0 unlabeled. Bounded (domains 1 to 4) and unbounded Z3 agree on all 389 equivalences.

The corrected gold turned out to be a story-level re-annotation on the Hugging Face Hub (DSAVlab-UNIUD/FOLIO_validation-curated). Its sentence text matches FOLIO v0.0 for only 97 of the 300 screen sentences, so the primary label uses the curated gold on those 97 and the original gold elsewhere. The curated release changed 50 of the 97 golds (52%).

The mutant generator kept 1,375 solver-verified non-equivalent mutants over 9 operators:

Operator | NEG | QUANT | IMPL_REV | DROP | ADD | ARG_SWAP | ANDOR | MERGE | CARD | SCOPE
kept | 300 | 157 | 124 | 95 | 300 | 102 | 109 | 151 | 37 | 0

It also kept 412 solver-verified equivalent rewrites: synonym renaming 142, contrapositive 115, De Morgan 80, reorder 71, prenex 4. The scope-swap operator applied to none of the 300 golds, because FOLIO-dev gold contains no mixed quantifier prefixes.

Complexity terciles were cut on a composite of token count, quantifier count, nesting depth and condition count. The top tercile holds 286 candidates, of which only 73 are equivalent to gold; the low tercile holds 264, of which 170 are equivalent.

Table 2. Label diagnostics per system (Arm A). The primary labels mix curated and original gold; the second column uses the original gold only.

System	n	equivalent (primary)	equivalent (original gold)	equivalent (lexical bijection)	parse rate
gpt-3.5-turbo	265	0.453	0.523	0.434	0.966
gpt-4	298	0.550	0.623	0.527	0.980
text-davinci-003	272	0.386	0.454	0.386	0.967

Among the non-equivalent labels, 352 are *vocabulary-shape mismatches*: the candidate's multiset of predicate arities differs from the gold's, so no bijection can exist. That is 79% of the non-equivalences, and it is where correct-but-gold-inequivalent candidates with a different granularity land. The other non-equivalences are 70 with a bijection but no equivalence, and 24 unparseable. Labels against the original and the curated gold agree on 75% of the 277 items that have both. Sixty-two items are equivalent to the original gold but not to the curated one, and 4 go the other way. The blind and lexical bijection labels disagree on 12 items. These are reversed implications that a blind predicate swap maps onto the gold.

Configuration decisions taken during execution

The probe configuration was chosen on 100 MED items by a pre-registered rule: keep thinking off unless its downward accuracy is below 0.80 and thinking raises it by at least 0.05. Table 3 gives the pilot.

Table 3. Substitution-probe pilot on 100 MED items. Both gemini configurations fell below the 0.70 downward threshold, which pre-registered a switch to gpt-4.1-mini. gpt-4.1-mini was worse, so the switch was not made.

Probe configuration | overall | upward | downward | downward, entailment | downward, neutral
gemini-2.5-flash, thinking off | 0.68 | 0.88 | 0.48 | 0.36 | 0.60
gemini-2.5-flash, thinking 1024 (chosen) | 0.77 | 0.90 | 0.64 | 0.52 | 0.76
gpt-4.1-mini, no thinking | 0.59 | 0.86 | 0.32 | 0.24 | 0.40

Before any real-candidate label was used, the text side was tuned on gold, mutant and rewrite behaviour only:

- optional domain nouns and light verbs (person, thing, have, take);
- predicates allowed to align to proper-noun concepts, and constants to noun concepts.

The share of golds with a full signature match rose from 0.31 to 0.43; the gold-signature oracle reaches 0.53. The residue is genuine granularity, because FOLIO golds omit content words the sentence contains.

The decomposed judge cost 3.8 times its estimate. Its mutant sample was cut to 20 per operator and its rewrites to 120, and a tuned-judge tier was skipped under the budget rules.

Main result on real candidates

Table 4 is the arm's headline table. On the full screen the three signature variants rank above every baseline except the decomposed judge and the oracle. All three have an increment over judge plus parse rate whose confidence interval excludes zero. In the top complexity tercile every metric falls to chance.

Table 4. Arm A: all metrics on 835 real candidates (389 equivalent / 446 not), $1000 \times$ sentence-clustered bootstrap. The false-alarm threshold is $\delta = 0$ for signature-like scores and $\delta = 0.10$ for continuous scores. " Δ best" is the increment over a logistic stack of judge, parse, round-trip NLI, structural metrics and self-consistency.

Metric	AUROC [95% CI]	low	mid	top tercile	G1 coverage	G2 false alarms	G3 Δ AUROC over judge+parse [CI]	Δ best [CI]	\$/item
A1 LLM-probe signature	0.711 [0.663, 0.756]	0.726	0.739	0.507	0.86	0.131	+0.084 [0.037, 0.138]	+0.057 [0.022, 0.092]	0.0009
A2 relativized signature	0.711 [0.662, 0.756]	0.724	0.739	0.507	0.86	0.131	+0.084 [0.036, 0.138]	+0.056 [0.021, 0.092]	0.0009
A3 rule-marker signature	0.718 [0.673, 0.763]	0.757	0.737	0.508	0.88	0.143	+0.093 [0.048, 0.143]	+0.062 [0.029, 0.097]	0
A0 alignment-only control	0.674 [0.629, 0.715]	0.694	0.665	0.519	0.86	0.133	+0.067 [0.025, 0.113]	+0.036 [0.008, 0.065]	0
Ccov concept-coverage control	0.661 [0.616, 0.706]	0.677	0.651	0.460	0.97	0.160	+0.056 [0.019, 0.099]	+0.024 [0.004, 0.044]	0
B1 whole-formula judge	0.644 [0.597, 0.693]	0.680	0.623	0.533	1.00	0.308	n/a	n/a	0.00002
B2 parse rate	0.527 [0.515, 0.541]	0.532	0.522	0.528	1.00	n/a	n/a	n/a	0
B2sat parse + satisfiable	0.529 [0.517, 0.543]	0.532	0.525	0.531	1.00	n/a	+0.000	[-0.000,	

0.001] | +0.000 [−0.001, 0.001] | 0

B3cos round-trip MiniLM | 0.614 [0.562, 0.663] | 0.702 / 0.517 / 0.424 | 1.00 | n/a | +0.015 [−0.008, 0.038] | +0.000 [−0.005, 0.005] | 0.00006

B3nli round-trip DeBERTa NLI | 0.660 [0.612, 0.706] | 0.681 / 0.625 / 0.472 | 1.00 | n/a | +0.027 [−0.005, 0.060] | +0.031 [0.001, 0.061] | 0.00006

B4 decomposed judge | 0.715 [0.665, 0.760] | 0.710 / 0.769 / 0.521 | 0.82 | 0.183 | +0.088 [0.040, 0.142] | +0.064 [0.027, 0.102] | 0.0041

B5 gold-signature oracle | 0.741 [0.694, 0.782] | 0.773 / 0.758 / 0.583 | 0.87 | 0.141 | +0.114 [0.065, 0.168] | +0.080 [0.043, 0.117] | n/a

B7 structural (user pilot) | 0.537 [0.519, 0.557] | 0.561 / 0.533 / 0.545 | 1.00 | n/a | −0.002 [−0.011, 0.006] | −0.002 [−0.006, 0.001] | 0

B8 self-consistency | 0.608 [0.564, 0.655] | 0.576 / 0.640 / 0.701 | 0.55 | n/a | +0.027 [−0.009, 0.066] | +0.028 [0.006, 0.051] | 0

The result does not hinge on the label choice. Table 5 re-scores the main metrics under the alternative labels. Every metric rises on the 97-sentence subset with curated gold. The ordering, with the signature above the judge, holds under every labelling.

Table 5. AUROC of Arm A metrics under alternative labellings. "Curated subset" has 277 candidates on the 97 sentences with curated gold; "unparseable as 0" scores unparseable candidates 0 instead of 0.5.

Metric | primary | original gold | lexical bijection | curated subset | unparseable as 0

A1 LLM-probe signature | 0.711 | 0.712 | 0.717 | 0.760 | 0.716

A3 rule-marker signature | 0.718 | 0.714 | 0.725 | 0.782 | 0.723

A0 alignment-only | 0.674 | 0.663 | 0.679 | 0.744 | 0.677

B1 judge | 0.644 | 0.666 | 0.649 | 0.721 | 0.659

B3nli round-trip | 0.660 | 0.654 | 0.656 | 0.755 | 0.675

B4 decomposed judge | 0.715 | 0.719 | 0.719 | 0.770 | 0.721

B5 oracle | 0.741 | 0.743 | 0.752 | 0.791 | 0.744

B8 self-consistency | 0.608 | 0.578 | 0.596 | 0.521 | 0.621

Invariance: the gate every signature variant failed

All three variants failed the false-alarm gate, and Table 6 shows where the failures come from. On the 270 purely logical rewrites (reorder, contrapositive, De Morgan, prenex) the signature metrics raised no alarms. Every alarm came from synonym-renamed predicates, where the lexical aligner lost the concept. The whole-formula judge alarmed on 20% to 40% of the logical rewrites.

Table 6. False-alarm rate on solver-verified equivalent rewrites, by rewrite kind (Arm A). The decomposed judge was run on a 120-rewrite subsample.

Rewrite kind | n | A1 | A3 | A0 | B5 oracle | B1 judge | B4 (n)

synonym rename | 142 | 0.380 | 0.415 | 0.387 | 0.408 | 0.246 | 0.311 (45)

contrapositive | 115 | 0.000 | 0.000 | 0.000 | 0.000 | 0.400 | 0.065 (31)

De Morgan | 80 | 0.000 | 0.000 | 0.000 | 0.000 | 0.388 | 0.211 (19)

reorder | 71 | 0.000 | 0.000 | 0.000 | 0.000 | 0.197 | 0.087 (23)

prenex | 4 | 0.000 | 0.000 | 0.000 | 0.000 | 0.250 | 0.000 (2)

all | 412 | 0.131 | 0.143 | 0.133 | 0.141 | 0.308 | 0.183 (120)

At $\delta = 0$ the judge's false-alarm rate is 0.43.

Verdict under the pre-registered rule. No signature variant survives. All pass the coverage gate,

the increment gate and the AUROC floor; all fail the false-alarm gate. Under the no-survivor clause the rule-marker signature advances, because its increment point estimate (+ 0.093) is the largest and exceeds 0.03.

Diagnosis: which half of the bridge failed

The design was meant to locate a failure, and it does. The text side is the bottleneck. On MED and HELP (Table 7) the LLM probe's downward accuracy is 0.60. That is well under the 0.80 bar the hypothesis set for itself and under the 0.85 it assumed. Downward entailments are the weak cell: 0.42 correct. The zero-LLM rule marker is scored on 3,401 MED single-span edits. That is a different item set from the LLM's 400, so the comparison is not exact, but the marker reaches 0.87 overall and 0.86 downward.

Table 7. Text-side accuracy on the monotonicity NLI sets (B6).

Set | n | overall | upward | downward | downward entailment | downward neutral

MED + HELP, LLM probe | 400 | 0.738 | 0.875 | 0.600 | 0.420 | 0.780

MED, LLM probe | 300 | 0.763 | 0.900 | 0.627 | 0.467 | 0.787

HELP, LLM probe | 100 | 0.660 | 0.800 | 0.520 | 0.280 | 0.760

MED single-span edits, rule marker | 3,401 | 0.870 | 0.890 (n=1,346) | 0.856 (n=2,055) | n/a | n/a

The same ordering shows up on the screen's own gold formulas. The text-side labels were compared with the gold formula's solver labels on the aligned concepts. The rule marker agreed on 87.2% of 499 concepts and the LLM probe on 78.3% of 442. Both do well on upward and downward concepts. On concepts the gold marks non-monotone, the LLM probe agreed on only 17.5% (n = 40) and the marker on 69.6% (n = 46).

The contrasts in Table 8 carry the rest of the diagnosis.

Table 8. Paired AUROC contrasts on real candidates (Arm A), sentence-clustered 95% CIs. The last four rows are increments over a logistic stack of [judge, parse_ok, alignment-only control].

Contrast | ΔAUROC [95% CI]

B5 oracle – A1 (text-side gap) | +0.030 [0.005, 0.058]

A1 – B4 (exact semantics vs decomposition, n = 810) | –0.005 [–0.024, 0.014]

A3 – A1 | +0.007 [–0.024, 0.039]

A1 – A0 (polarity over alignment) | +0.037 [0.006, 0.069]

A2 – A0 | +0.036 [0.006, 0.068]

A3 – A0 | +0.044 [0.009, 0.078]

B4 – A0 | +0.041 [0.007, 0.074]

A1 – Ccov | +0.050 [0.013, 0.082]

A3 – Ccov | +0.057 [0.030, 0.084]

A1 over [B1, parse, A0] | +0.013 [–0.013, 0.040]

A2 over [B1, parse, A0] | +0.013 [–0.013, 0.039]

A3 over [B1, parse, A0] | +0.024 [–0.005, 0.056]

B4 over [B1, parse, A0] | +0.017 [–0.012, 0.047]

Three readings follow.

1. **The oracle gap is small.** A perfect text side would add only 0.030 AUROC, so even perfect probes would leave the signature near 0.74 on these labels.
2. **Exact semantics buys nothing on real items.** Solver-exact formula semantics ties the decomposed LLM judge, at about a fifth of the judge's cost. The two scores are highly

correlated (Spearman 0.917).

3. **Much of the signal is lexical alignment.** The alignment-only control alone reaches 0.674. Standalone, the polarity comparison adds 0.04 over it. Once the judge and the alignment-only control are both in the model, the polarity increment is no longer significant for any variant.

The median alignment coverage on real candidates was 0.8, so alignment failure was not the limiting factor for coverage.

Complexity: the predicted crossover did not appear

The hypothesis predicted that the signature would overtake the judge as the number of conditions grows. Table 9 shows the opposite pattern on FOLIO, and Figure 1 (Section 1.4) sets it beside the Arm B metrics. The signature's lead over the judge is largest in the middle tercile and gone in the top tercile, where every metric is close to chance. The one metric that rises with complexity is self-consistency, and only on its 55% covered subset.

Table 9. Difference from the whole-formula judge, by complexity tercile (Arm A), with 95% CIs. Terciles: low n = 264 (170 equivalent), mid n = 285 (146), top n = 286 (73).

Metric | low | mid | top | slope across terciles [CI]

A1 – B1 | +0.046 [–0.049, 0.144] | +0.116 [0.015, 0.216] | –0.027 [–0.145, 0.092] | –0.036 [–0.115, 0.044]

A3 – B1 | +0.077 [–0.016, 0.174] | +0.114 [0.014, 0.213] | –0.026 [–0.144, 0.098] | –0.051 [–0.128, 0.028]

B4 – B1 | +0.029 [–0.063, 0.128] | +0.146 [0.034, 0.254] | –0.013 [–0.129, 0.111] | –0.021 [–0.094, 0.059]

B5 – B1 | +0.093 [0.004, 0.194] | +0.135 [0.038, 0.235] | +0.050 [–0.059, 0.153] | –0.022 [–0.093, 0.053]

B8 – B1 | –0.104 [–0.209, –0.001] | +0.018 [–0.084, 0.124] | +0.168 [0.046, 0.294] | +0.136 [0.058, 0.217]

The arm's analysis attached a caveat that the dataset artifact later confirmed: FOLIO's top tercile is only mildly conditioned. FOLIO contains a single "unless"/"except" in 2,656 sentence-formula pairs, so the crossover test was weak by construction.

Controlled mutants: mechanism, blind spots and error typing

The per-operator results in Table 10 are pairwise detection rates: the probability that a mutant scores below its gold, with ties counted as one half. On mutants the whole-formula judge beats every signature variant on every operator, often by a wide margin. The pre-registered blind spots came true. Cardinality mutants sit at exactly 0.50 for every signature metric, and so do conjunction/disjunction swaps. The relativized signature did not rescue conjunction/disjunction swaps (0.50), although its formula side separates them in unit tests with at least 6 differing coordinates. The text side produced relativized coordinates for very few FOLIO sentences.

Table 10. Pairwise mutant detection (tie = 0.5) by operator, Arm A. The decomposed judge (B4) used 20 mutants per operator; all other rows use every kept mutant (n in the header).

Metric | NEG (300) | QUANT (157) | IMPL_REV (124) | DROP (95) | ADD (300) | ARG_SWAP (102) | ANDOR (109) | MERGE (151) | CARD (37)

A1 LLM-probe signature | 0.758 | 0.720 | 0.782 | 0.742 | 0.802 | 0.613 | 0.495 | 0.818 | 0.500

A2 relativized | 0.758 | 0.720 | 0.782 | 0.742 | 0.802 | 0.613 | 0.500 | 0.818 | 0.500

A3 rule-marker | 0.808 | 0.726 | 0.770 | 0.805 | 0.823 | 0.613 | 0.518 | 0.838 | 0.500
A0 alignment-only | 0.500 | 0.500 | 0.500 | 0.742 | 0.798 | 0.500 | 0.500 | 0.844 | 0.500
Ccov coverage | 0.500 | 0.500 | 0.500 | 0.842 | 0.475 | 0.500 | 0.500 | 0.950 | 0.500
B5 gold-signature oracle | 0.848 | 0.809 | 0.879 | 0.763 | 0.828 | 0.691 | 0.528 | 0.871 | 0.500
B1 whole-formula judge | 0.895 | 0.822 | 0.899 | 0.784 | 0.877 | 0.647 | 0.862 | 0.871 | 0.919
B4 decomposed judge (n=20 each) | 0.650 | 0.650 | 0.700 | 0.725 | 0.725 | 0.500 | 0.600 | 0.750 | 0.475

[Figure fig2: generated in the report phase, not in this draft]

For the LLM-probe signature, the confidence intervals for the blind spots were:

- cardinality: 0.50 [0.35, 0.65], n = 37;
- conjunction/disjunction: 0.50 [0.40, 0.59], n = 109.

Scope swaps could not be tested, because no gold had mixed quantifiers. Ties are frequent. Counting only mutants scored strictly below their gold, the LLM-probe signature detects negation 0.573, $\forall/\exists$ 0.490, argument swap 0.304 and conjunction/disjunction 0.037, against 0.797, 0.701, 0.461 and 0.780 for the whole-formula judge. Scoring on raw mismatch fractions, without the 0.5 given to uncovered items, gives slightly higher values for the LLM-probe signature: negation 0.785, implication reversal 0.823, merge 0.940.

Error typing was weak. Macro recall over the 8 non-blind error types was 0.52 for the LLM-probe signature, 0.54 for the relativized signature and 0.56 for the rule marker, against 0.125 for chance. Argument swaps were always typed correctly: 31 of the 31 detected. Negation errors were often typed as conflation, for example 48 of 141 for the LLM-probe signature. Implication reversals were typed as conflation 27 times and correctly 26 times. Table 11 gives the full confusion counts for the rule-marker signature, the variant that advanced.

Table 11. Error-type confusion for the rule-marker signature (A3) on detected mutants; rows are the true operator family, columns the predicted type.

True \ predicted	negation	quantifier	reversed impl.	added cond.	dropped cond.	swapped args	conflation	and/or	other
negation/polarity	75	34	0	5	0	5	78	0	8
quantifier	8	39	2	2	0	1	31	0	5
reversed implication	0	9	30	1	0	1	33	0	0
added condition	3	0	0	112	68	11	21	0	0
dropped condition	0	1	0	0	52	2	9	0	1
swapped args	0	0	0	0	0	31	0	0	0
conflation (merge)	0	0	0	0	37	1	72	0	0
and/or	0	0	0	1	0	0	4	2	0

System level, gold-error flagging, correlations, cost

With three systems, Kendall τ can take only four values, so it was reported but not read. Every metric put text-davinci-003 last, and the metrics disagreed on the order of gpt-3.5-turbo and gpt-4. τ was 1.0 for the judge, parse-plus-satisfiable, round-trip cosine, the structural metrics and self-consistency, and 0.33 for all signature variants. The signature variants' system means are gpt-3.5-turbo 0.774, gpt-4 0.762 and text-davinci-003 0.701 for the LLM-probe signature, against label accuracies of 0.453, 0.550 and 0.386.

Gold-error flagging could be tested only on the 72 original golds whose sentences match the curated release (Table 12). It is weak: precision at 50 is barely above the base rate.

Table 12. Flagging wrong gold without reference: signature score of the original gold vs the curated release's "corrected" flag, on 72 golds with a base rate of 0.583.

Metric | precision@50 | AUROC of flag

A1 LLM-probe signature | 0.62 | 0.623

A2 relativized | 0.62 | 0.623

A3 rule-marker | 0.64 | 0.542

The signature variants correlate strongly with the decomposed judge (Spearman 0.917 for the LLM-probe signature, 0.719 for the rule marker), the oracle (0.856 and 0.787) and the alignment-only control (0.798 and 0.738). They correlate weakly with the whole-formula judge (0.374 and 0.370), round-trip NLI (0.464 and 0.448), parse rate (about 0.20) and self-consistency (0.082 and 0.113). The complementarity with the judge is what produces the increment in Table 4.

Total spend on OpenRouter was \$5.67 against a \$5.80 cap, over 5,504 calls:

Purpose | \$

decomposed judge (B4) | 4.719

substitution probe (A1/A2) | 0.747

whole-formula judge (B1) | 0.071

MED/HELP evaluation (B6, full) | 0.051

back-translation (B3) | 0.049

probe-configuration pilots | 0.018

decomposed-judge test and thinking check | 0.010

Amortized per candidate, the LLM-probe signature costs \$0.0009 (the probe is \$0.0025 per sentence and is shared by all candidates of that sentence), the rule marker \$0, and the decomposed judge \$0.0041. Solver time is about 0.016 s per item; the judge takes 0.56 s.

A separate script (`audit_rederive.py`) re-derived the headline numbers from the raw per-item files, with no code shared with the analysis. It used a hand-written Mann-Whitney AUROC, its own 500-draw clustered bootstrap, and a Newton-Raphson logistic regression with hash-based folds. All AUROCs matched and the CIs agreed within ± 0.005 . The increment estimates differed by at most 0.006: the LLM-probe signature +0.078 [0.031, 0.124], the rule marker +0.090 [0.046, 0.140]. Placebos behaved as required. With permuted labels the rule marker's AUROC was 0.528 and its increment CI $[-0.009, 0.059]$ included zero. A random-noise metric scored 0.486 with increment $-0.002 [-0.012, 0.008]$. For this report we recomputed the Arm A AUROCs once more from `results/screen_scores.jsonl` and `data/screen_set.json` (`checks/recompute_auroc.py`), and every value in Table 4 reproduced to three decimals.

1.4 Screen Arm B: solver worlds, instance consequences and cross-system agreement

What was built

Arm B tested the three alternates against the same judge prompt.

- **Truth-value-judgment worlds (TVJT).** The test is borrowed from child-language research on truth-value judgment and from the distinguishing-input idea of oracle-guided program selection [24]. For a candidate F and each of its own typed mutants M , Z3 finds a minimal world (domain size ≤ 4) where F and M disagree. The world is verbalized as a closed list of facts ("These are ALL the facts"). In one batched call, gemini-2.5-flash (temperature 0, no

reasoning) judges whether the *sentence* is true in each world; it never sees F or M. The score is the share of worlds where the judged truth equals F's truth, and the predicted error type is the operator whose worlds F loses most.

- **Instance-consequence NLI.** Z3 labels premise-hypothesis pairs about named individuals as entailed, contradicted or neutral under F. Hypotheses include ground literals and templated quantified consequences: shared witness, existence, universal, at-least-two. DeBERTa-v3-large NLI [27] reads *sentence* + *facts* $\Rightarrow$ *hypothesis*, with zero LLM calls. The score is the mean agreement with F's labels.
- **One-coin latent-class model.** Per sentence, the three systems' outputs are clustered by blind bijection equivalence. A one-coin latent-class EM [22] with an explicit all-wrong state gives the posterior that each candidate's class is the correct one. A two-population variant in the style of Hui and Walter [23] fits accuracies per complexity tercile. The gold is excluded, to avoid circularity.

The baselines were the same whole-formula judge (identical prompt string), parse rate, and a zero-cost self-consistency score. Secondary variants were:

- TVJT with Logic-LM's predicate glosses in the verbalizer;
- TVJT restricted to non-vacuous worlds, with at least one instance of every universal restrictor;
- the same NLI pairs judged by gemini;
- majority agreement;
- a binary Dawid-Skene model;
- latent-class clustering under a trigram-anchored predicate map.

Screen set and label policy as built by Arm B

The recipe yielded 300 sentences and 833 real candidates: gpt-3.5-turbo 265, gpt-4 297, text-davinci-003 271. All three systems translated 246 of the sentences. Fifty-eight candidates (7.0%) were unparseable: free variables 34, non-FOL symbols 15, function terms 8, tokenizer failure 1. They were kept, labelled incorrect and scored 0.5 by every metric.

Arm B departed from Arm A on the primary gold, and recorded why. The curated release rewords story sentences, uses binary predicates and constants where FOLIO v1 used unary ones, and adopts different conventions ("some" = at least two). Equivalence against it therefore mostly measures vocabulary mismatch. Arm B took FOLIO v1 gold as primary (usable on 299 of 300 sentences) and the curated gold as a secondary label. The primary label split is 463 correct / 370 incorrect / 0 unlabeled, and all 463 bounded equivalences were confirmed unbounded.

Arm B also found that Arm A zips premise lists even when the premise and premise-FOL counts differ, which misaligns gold in stories 87, 105, 106 and 173. The two screens therefore differ on about 11 sentences. The 803 shared items carry identical candidates.

Arm B quantified the label noise in its own terms:

- **Wrong gold.** On 108 sentences with parseable curated gold, v1 gold is bijection-equivalent to the curated gold on 65 and non-equivalent on 43. Thirty-seven of the 43 are vocabulary or granularity differences; only 6 (5.6%) are semantic disagreements with compatible vocabulary. The gold-as-rater latent-class model puts v1 gold error at 22.9%. The observed curated-changed rate is 33.9% on 109 sentences.

- **Correct but not equivalent.** 250 of the 370 "incorrect" labels (67.6%) are vocabulary mismatches: a different arity multiset or constant count. Such candidates can never be bijection-equivalent, so this share upper-bounds that noise. Switching from v1 to curated gold flips 64 v1-correct items to incorrect and only 9 the other way, a 23.9% disagreement on 305 items. The remaining incorrect labels are 62 with no equivalent map and 58 unparseable.
- **Blind-bijection false positives.** The blind and trigram-anchored labels disagree on 2.0% of items (17 are correct under the blind map only). The equivalence map is not unique for 11.6% of items.

Main result on real candidates

Table 13 is Arm B's headline table. It adds a within-sentence accuracy, which removes between-sentence difficulty: the probability that a correct candidate outcores an incorrect one for the same sentence, over the 94 sentences with mixed labels.

Table 13. Arm B: all metrics on 833 real candidates (463 correct / 370 not), $1000 \times$ sentence-clustered CIs, seed 0. The false-alarm column gives the rate at the Youden threshold τ^* , then at 0.5. "Vocab-compatible" restricts to items whose arity profile could match the gold.

Metric	role	AUROC [95% CI]	tercile low / mid / top	G1 coverage	G2 FA@ τ^* (FA@0.5)	G3 Δ AUROC [CI]	within-sentence [CI]	AUROC vocab-compatible	\$/item	s/item (median)
TVJT	candidate	0.701 [0.656, 0.745]	0.669 / 0.684 / 0.751	0.930	0.479 (0.184)	+0.039 [+0.011, +0.067]	0.729 [0.666, 0.791]	0.706	0.000191	0.69
Instance NLI (DeBERTa)	candidate	0.603 [0.550, 0.647]	0.766 / 0.450 / 0.518	0.918	0.683 (0.028)	+0.001 [−0.017, +0.018]	0.537 [0.444, 0.626]	0.571	0	0.06
One-coin latent class	candidate	0.853 [0.813, 0.888]	0.863 / 0.836 / 0.829	0.900	n/a	+0.111 [+0.073, +0.150]	0.765 [0.681, 0.839]	0.814	0	0.00
B1 whole-formula judge	baseline	0.665 [0.616, 0.711]	0.718 / 0.646 / 0.552	1.000	0.527 (0.409)	n/a	0.593 [0.514, 0.671]	0.663	0.000024	0.51
B2 parse rate	baseline	0.578 [0.558, 0.600]	0.569 / 0.530 / 0.612	1.000	0.000 (0.000)	n/a	0.550 [0.550, 0.624]	0.500	0	0.00
B3sc self-consistency	baseline	0.597 [0.553, 0.642]	0.568 / 0.613 / 0.703	0.519	n/a	−0.013 [−0.050, +0.023]	0.610 [0.556, 0.664]	0.539	0	0.00
TVJT with glosses	secondary	0.716 [0.680, 0.752]	0.732 / 0.661 / 0.718	0.930	n/a	+0.044 [+0.016, +0.075]	0.740 [0.681, 0.796]	0.696	0.000174	0.67
TVJT non-vacuous worlds	secondary	0.667 [0.621, 0.709]	0.669 / 0.672 / 0.637	0.930	0.489 (0.238)	+0.022 [−0.004, +0.048]	0.652 [0.581, 0.728]	0.670	0.000196	0.69
Instance NLI with glosses	secondary	0.610 [0.561, 0.658]	0.770 / 0.474 / 0.516	0.918	n/a	−0.009 [−0.031, +0.010]	0.548 [0.461, 0.638]	0.571	0	0.06
Instance NLI judged by gemini	secondary	0.682 [0.645, 0.719]	0.771 / 0.655 / 0.659	0.918	n/a	+0.023 [+0.004, +0.044]	0.667 [0.599, 0.733]	0.618	0.000482	0.96
Hui-Walter latent class	secondary	0.855 [0.814, 0.890]	0.863 / 0.828 / 0.841	0.900	n/a	+0.110 [+0.072, +0.149]	0.754 [0.670, 0.831]	0.827	0	0.00
Majority agreement	secondary	0.821 [0.783, 0.854]	0.837 / 0.815 / 0.765	0.900	n/a	+0.158 [+0.115, +0.203]	0.650 [0.575, 0.720]	0.798	0	0.00
Dawid-Skene (binary)	secondary	0.833 [0.793, 0.866]	0.849 / 0.837 / 0.759	0.900	n/a	+0.168 [+0.122, +0.214]	0.630 [0.542, 0.707]	0.816	0	0.00
One-coin, trigram clustering	secondary	0.853 [0.813, 0.886]	0.869 / 0.836 / 0.829	0.900	n/a	+0.116 [+0.077, +0.159]	0.765 [0.681, 0.839]	0.811	0	0.00

The paired contrasts in Table 14 carry the arm's two substantive findings. Cross-system agreement is by far the strongest signal. TVJT is the only text-reading metric in the whole screen whose advantage over the judge grows with complexity: +0.199 in the top tercile, where the judge falls to 0.552.

Table 14. Paired AUROC differences on identical items (Arm B), sentence-clustered 95% CIs.

Comparison | Δ AUROC, all items | Δ AUROC, top tercile

TVJT – B1 judge | +0.036 [–0.015, +0.094] | +0.199 [+0.085, +0.308]
one-coin – majority agreement | +0.032 [+0.018, +0.047] | +0.064 [+0.028, +0.104]
one-coin – B1 judge | +0.188 [+0.137, +0.240] | +0.277 [+0.159, +0.388]
TVJT with glosses – TVJT | +0.015 [–0.021, +0.053] | –0.033 [–0.088, +0.022]
TVJT non-vacuous – TVJT | –0.035 [–0.058, –0.013] | –0.114 [–0.180, –0.046]
instance NLI – B1 judge | –0.062 [–0.117, –0.010] | –0.034 [–0.121, +0.046]
NLI judged by gemini – DeBERTa NLI | +0.079 [+0.030, +0.129] | +0.141 [+0.070, +0.212]
TVJT – NLI judged by gemini | +0.019 [–0.026, +0.063] | +0.091 [+0.025, +0.168]

[Figure fig1: generated in the report phase, not in this draft]

Verdict under the pre-registered rule. The one-coin latent-class model is the only survivor and the rule's winner. It clears coverage (0.900), the increment gate (+0.111 [0.073, 0.150]) and the AUROC floor (0.853); the false-alarm gate does not apply to it, because it scores sets of outputs, not single formulas. TVJT clears coverage (0.930), the increment gate (+0.039 [0.011, 0.067]) and the AUROC floor (0.701), and fails only the false-alarm gate. Its rewrites draw different worlds from their own mutants, and the judge disagrees on some of those worlds. The false-alarm rate by rewrite kind at τ^* was:

- rename 0.584;
- reorder 0.424;
- De Morgan 0.346;
- contrapositive 0.308.

Instance-consequence NLI with DeBERTa fails the increment gate (+0.001), the AUROC floor (0.603) and the false-alarm gate (0.683).

The arm recorded one caveat about its winner. The labels and the agreement use the same equivalence machinery: if two systems agree and one is gold-equivalent, the other is too, so agreement is favoured by construction. A second caveat comes from the epidemiology the model borrows: latent-class estimates are biased when the raters' errors are conditionally dependent [36], and three systems of one era, prompted by one pipeline, cannot test that assumption. The model also needs at least two systems and cannot score a lone output. At system level it orders the systems gpt-3.5-turbo, gpt-4, text-davinci-003; the labels order them gpt-4, gpt-3.5-turbo, text-davinci-003. Table 15 gives the fitted parameters.

Table 15. Fitted latent-class parameters (Arm B). p_w is the estimated per-system accuracy; the gold-as-rater fit is used only to estimate gold error and gives no AUROC, since gold is the label source.

Model | gpt-3.5-turbo | gpt-4 | text-davinci-003 | other

one-coin p_w | 0.771 | 0.786 | 0.622 | $\pi = 0.985, \rho = 0.001$, 18 iterations
Hui-Walter p_w , low tercile | 0.876 | 0.879 | 0.721 | $\pi = 0.990$
Hui-Walter p_w , mid tercile | 0.885 | 0.778 | 0.694 | $\pi = 0.999$

Hui-Walter p_w, top tercile | 0.550 | 0.726 | 0.449 | π = 0.958
gold as 4th rater p_w | 0.693 | 0.812 | 0.569 | p_gold = 0.771, estimated gold error 0.229

Controlled mutants and the blind-spot head-to-head

On the 300 screen golds Arm B kept 1,187 mutants and 499 rewrites (RENAME 274, CONTRAPOSITIVE 104, REORDER 66, DEMORGAN 55). Its mutant counts differ from Arm A's because the operators were implemented separately. Table 16 gives TVJT pairwise detection and standalone detection, the latter the share of mutants scored below τ^* . TVJT sees every operator family, including conjunction/disjunction swaps and cardinality, the two where the signature sits at 0.50.

Table 16. Per-operator detection on screen-gold mutants (Arm B). Pairwise = the judge sides with the gold on the separating world; standalone = mutant score < τ^* .

Operator | n | TVJT pairwise | NLI (DeBERTa) pairwise (undistinguishable) | TVJT standalone | B1 standalone

NEG | 300 | 0.838 | 0.788 (53) | 0.957 | 0.990
QUANT | 141 | 0.883 | 0.709 (12) | 0.950 | 0.957
IMPL_REV | 121 | 0.864 | 0.703 (30) | 0.984 | 0.992
DROP_CONJ | 82 | 0.860 | 0.677 (35) | 0.744 | 0.671
ADD_CONJ | 300 | 0.863 | 0.582 (213) | 0.890 | 0.803
ARG_SWAP | 81 | 0.883 | 0.735 (27) | 0.901 | 0.407
AND_OR | 94 | 0.830 | 0.702 (22) | 0.957 | 0.947
SCOPE_SWAP | 0 | n/a | n/a | n/a | n/a
MERGE | 40 | 0.825 | 0.713 (13) | 0.325 | 0.575
CARD | 28 | 0.762 | 0.679 (14) | 0.929 | 0.893

The judge's accuracy on worlds drawn from the gold formulas was 0.856. The DeBERTa model scored the gold formulas' own consequences at 0.820 mean agreement, and gemini at 0.939. Error-type identification was poor: macro-F1 over 10 operator classes was 0.253 for TVJT (top-1 0.356) and 0.181 for NLI (top-1 0.212). Additions of a fresh predicate on ground formulas are invisible to instance NLI, because its hypotheses use only the candidate's own vocabulary; this is why 213 of 300 addition mutants were undistinguishable.

FOLIO-dev gold has no mixed $\forall/\exists$ formulas, so scope swaps could not be generated on the screen. Arm B built a blind-spot supplement from the remaining FOLIO-dev golds, FOLIO-train, curated MALLS and MALLS-v0.1-test (37 scope swaps, 440 cardinality mutants), used only for this head-to-head (Table 17). TVJT detects scope swaps above chance; DeBERTa NLI does not.

Table 17. Blind-spot head-to-head: detection of scope and cardinality mutants, with Wilson 95% intervals. The signature was predicted, and measured in Arm A, to sit at 0.50 on these operators.

Metric | operator | split | detection | n | Wilson 95%

TVJT | SCOPE_SWAP | screen + supplement | 0.649 | 37 | [0.488, 0.782]
TVJT | CARD | screen | 0.762 | 84 | [0.661, 0.840]
TVJT | CARD | screen + supplement | 0.662 | 468 | [0.618, 0.704]
NLI (DeBERTa) | SCOPE_SWAP | screen + supplement | 0.473 | 37 | [0.322, 0.629]
NLI (DeBERTa) | CARD | screen | 0.643 | 56 | [0.512, 0.756]
NLI (DeBERTa) | CARD | screen + supplement | 0.605 | 440 | [0.558, 0.649]

Coverage, cost, reproducibility

Uncovered items received 0.5 and stayed in every AUROC. Coverage was TVJT 0.930, NLI 0.918, latent class 0.900 and self-consistency 0.519. World search covered every parseable target, with 4.27 worlds per target on average. The verbalizer failed on 7.0% of items and 4.0% of predicates. Per item, TVJT costs \$0.000191 and 0.69 s, the judge \$0.000024 and 0.51 s, gemini NLI \$0.000482 and 0.96 s, and DeBERTa NLI \$0 and 0.06 s. The latent class model's pairwise equivalence checks ran in the build stage. Total OpenRouter spend was \$1.55 against a \$6.50 hard cap. On a 20-sentence development slice outside the screen, the TVJT prompt reached a judge accuracy of 0.702 with no malformed JSON, and was then frozen. On 8 blind-spot probe pairs the judge sided with the gold on 7.

gemini-2.5-flash at temperature 0 with reasoning off was not deterministic. Identical prompts sent concurrently, for example when two systems emitted the same formula string, sometimes got different answers. The final numbers come from a second, cache-consistent pass, and the verdict did not change; TVJT moved by at most 0.002 AUROC.

`audit/rederive.py` re-derived the headline numbers through a different code path: a library AUROC, a separate cluster bootstrap, out-of-fold predictions from a separate cross-validation, brute-force within-sentence pairs and a direct majority recount. Point estimates matched exactly and CIs differed only in the third decimal. With permuted labels the AUROCs were 0.51 to 0.52 and the latent-class increment CI was $[-0.032, +0.043]$. A pure-noise metric scored 0.494 with increment CI $[-0.014, +0.010]$. The Z3 labels themselves and the EM posteriors were not independently re-derived. Our recomputation from `results/screen_scores.jsonl` reproduced every AUROC in Table 13 that we checked (one-coin 0.853, TVJT 0.701, DeBERTa NLI 0.603, judge 0.665, parse 0.578, self-consistency 0.597, gemini NLI 0.682, non-vacuous TVJT 0.667, majority 0.821).

1.5 A cross-arm check run for this report

The two arms were designed to rank all candidates on one set, but they ended up with different primary labels. On 803 shared items, Arm A's labels mark 382 correct (curated gold on 97 sentences, original gold elsewhere) and Arm B's mark 443 (FOLIO v1 gold). The two labellings agree on 90.7% of items. To see whether the ranking across arms depends on this, we scored every metric on the 803 shared items under both labellings (`checks/cross_arm.py`, output `checks/cross_arm_out.json`). This check was not part of either artifact and was not pre-registered.

Table 18. AUROC on the 803 items both arms share, under each arm's primary labels.

Metric (arm) | Arm A labels (382 correct) | Arm B labels (443 correct)

A1 LLM-probe signature (A) | 0.708 | 0.729
A3 rule-marker signature (A) | 0.720 | 0.729
A0 alignment-only control (A) | 0.667 | 0.674
B4 decomposed judge (A) | 0.711 | 0.737
B5 gold-signature oracle (A) | 0.733 | 0.752
B1 judge, Arm A run (A) | 0.636 | 0.665
B1 judge, Arm B run (B) | 0.643 | 0.666
B3nli round-trip (A) | 0.658 | 0.657
B8 self-consistency (A) | 0.619 | 0.581
One-coin latent class (B) | 0.772 | 0.849
TVJT (B) | 0.682 | 0.705
Instance NLI, DeBERTa (B) | 0.576 | 0.600
B3sc self-consistency (B) | 0.622 | 0.592

The ordering holds under both labellings: latent class, then the signature and decomposed judge, then TVJT, then the judge. The latent-class model loses 0.077 when the labels switch to Arm A's, far more than any other metric; the signature variants lose 0.01 to 0.02. The latent-class model's lead is therefore partly tied to the FOLIO v1 labelling, which fits the circularity caveat Arm B raised. The two independent runs of the judge prompt agree closely (0.636 and 0.643 under the same labels), so the judge anchor is stable across arms despite the non-determinism noted above.

1.6 The held-out meta-evaluation dataset

Why and what

The dataset artifact built the population on which iteration 2 is to confirm whatever the screen selects. It had to shift system era and corpus, oversample complex sentences, and carry labels that do not rest on solver equivalence alone. It holds 15,606 rows in five groups (Table 19). `data.py` re-verifies every row against the raw source files, and all 15,606 pass.

Table 19. Groups of the held-out dataset.

Group | Rows | Content

held-out confirmation | 6,300 | 700 sentences $\times$ 9 systems, greedy ($T = 0$); 301 unparseable outputs kept
held-out samples | 7,000 | 700 $\times$ 5 samples ($T = 0.8$) $\times$ {gpt-4.1-mini, llama-3.1-8b}; solver labels only
screen gold audit | 1,003 | the sha1-first 360 FOLIO-v1-validation sentences $\times$ 3 Logic-LM systems, labelled against panel-audited gold

contamination | 936 | 104 verified entity-renamed paraphrases $\times$ 9 renamed greedy candidates

transfer (unlabeled) | 367 | the user's EU-AI-Act Article 3 definitions $\times$ {grounded, ungrounded} $\times$ runs

The 700 sentences are:

- 250 from MALLS-v0.1-test, including all 99 eligible items the 2606.02837 authors corrected;
- 250 from FOLIO-v2-train, drawn from stories disjoint from both FOLIO validation versions (5 stories excluded, overlap asserted 0);
- 120 ProverQA-dev hard and 80 medium, de-templated, at most 2 per story.

45.1% fall in the top complexity tercile, whose cut-points were set on a 4,527-sentence eligible pool. The nine systems are llama-3.1-8b, qwen-2.5-7b, mistral-small-3.2-24b, gpt-4.1-mini, gemini-2.5-flash (no thinking), deepseek-v3.1, gemma-3-27b, phi-4 and gpt-oss-120b. They cover 7 families, from 8B models to frontier-mini ones, and all used one zero-shot prompt with free predicate naming.

Labels

Four label layers were built:

- **L0, gold audit.** A cascaded, family-disjoint three-model panel audited the gold.
- **L1, solver equivalence.** Lexical-free Z3 equivalence up to an arity-preserving bijection. Random-structure refutation comes first, then propositional grounding over domains 1 to 4, then an unbounded check. Only countermodels count as proof of non-equivalence.
- **L2, granularity-aware equivalence.** WordNet-lemma compound definitions of at most two literals, flagged as lexical.
- **L3, blinded adjudication.** The panel sees gold and candidate as Formula A and Formula B in random order, with no source named. 610 greedy items were adjudicated: 460 in a

stratified design plus 150 extra top-tercile items. Estimates use post-stratified weights.

L1 was validated on 59 parseable pairs among the first 60 MALLS items of the relabelling study. Every uncorrected pair was equivalent (37/37) and every corrected pair was non-equivalent (22/22).

Every panel member had to pass a calibration gate: at least 80% accuracy on 40 items with known labels. The gate was fixed before the members were chosen. The planned claude-sonnet-4.5 failed it (Table 20) because it rejected valid contrapositive and material-implication rewrites, so the pre-planned fallback was used.

Table 20. Panel calibration on 40 known-label items (gold-audit prompt; ≥ 0.80 required).

Member | Model | audit accuracy | adjudication accuracy | same-meaning accuracy | used

M1 (planned) | claude-sonnet-4.5 | 0.775 | 0.98 | 1.00 | no, failed gate

M1 | claude-haiku-4.5 | 0.900 | 0.90 | 0.84 | yes

M2 | grok-4.3, low reasoning | 0.900 | 0.98 | 1.00 | yes

M3 | glm-4.6, low reasoning | 0.925 | 0.90 | 0.92 | variant not used

M3 | glm-4.6, non-thinking | 0.875 | 0.94 | 0.88 | yes (cost)

Three of the 15 "faithful-by-construction" ProverQA-easy calibration golds drop a restrictor the sentence states ("For all fish" with no `fish(x)`), and all members flagged them. Gate accuracies were computed against the original labels regardless.

What the labels say about the benchmarks

Table 21 gives the audited wrong-gold rates. More than 40% of shipped gold is unfaithful on every corpus except ProverQA. Even ProverQA, a synthetic prover-generated benchmark, has 17.5% unfaithful gold, mostly dropped restrictors.

Table 21. Share of shipped gold the panel judged unfaithful (L0), with 95% CIs.

Corpus | n | wrong-gold rate [95% CI] | most frequent primary error

MALLS-v0.1-test | 250 | 0.456 [0.395, 0.518] | added condition 34, implication direction / "only" 28

FOLIO-v2-train (non-trivial) | 250 | 0.620 [0.558, 0.678] | dropped condition 35, added condition 30

FOLIO-v1-validation (screen, 360) | 360 | 0.442 [0.391, 0.493] | dropped condition 62, added condition 26

ProverQA-dev (all 200 audited) | 200 | 0.175 [0.129, 0.234] | dropped condition (16 of 35)

On the 99 MALLS items with a human correction, the panel agreed with the humans on 78.8% (Cohen $\kappa = 0.57$). Both judged the gold fine on 45 items and both judged it wrong on 33; the panel alone flagged 13 and the humans alone 8. The human wrong rate is 41.4% and the panel's 46.5%. On FOLIO the panel-vs-human check covers only 22 screen conclusions whose v1 gold equals the gold the paper judged: $\kappa = 0.18$, too few items to conclude anything. A second corrected FOLIO, yfxiao/folio-refined, barely agrees with the panel ($\kappa = 0.004$ on 416 items). Its undocumented refinement targets something other than faithfulness, so it was kept as metadata only. Agreement statistics for the panel were:

- L0 on the full-panel 20% subset: Fleiss $\kappa = 0.55$ (224 items);
- L3 on the candidate: $\kappa = 0.63$ (609 items, observed agreement 0.817);
- L3 on the gold: $\kappa = 0.54$.

Pairwise member κ ranged from 0.19 (haiku-grok) to 0.50 (grok-glm).

The central finding for the screen concerns the labels themselves (Table 22). Among candidates

that the solver called non-equivalent to the audited gold, the panel judged 43.9% faithful. On FOLIO the figure is 59.7%. Solver-only labels agree with the panel majority on 67.6% [63.5, 71.1] of the weighted frame. In the other direction, 9.9% of solver-equivalent candidates were judged unfaithful, because the gold they match is itself wrong. Of the 77 candidates that became equivalent only under the WordNet granularity definitions, 87.5% of the 40 adjudicated were judged faithful.

Table 22. Correct-but-gold-inequivalent rate: panel-faithful share (post-stratified weights) among solver-non-equivalent greedy candidates, with bootstrap 95% CIs.

Subset n adjudicated panel-faithful share [95% CI]
all non-equivalent to audited gold 546 0.439 [0.395, 0.486]
no bijection possible (arity mismatch) 472 0.447 [0.401, 0.496]
bijection exists, not equivalent 73 0.395 [0.288, 0.503]
non-equivalent to original gold 535 0.478 [0.436, 0.519]
equivalent only under granularity definitions 40 0.875 [0.775, 0.975]
FOLIO 224 0.597 [0.526, 0.664]
MALLS 205 0.327 [0.259, 0.400]
ProverQA 117 0.310 [0.233, 0.391]
bottom tercile 80 0.399 [0.288, 0.499]
middle tercile 95 0.484 [0.391, 0.577]
top tercile 371 0.429 [0.380, 0.477]

The adjudicated error-type distribution answers a question the hypothesis had left open: how much of what current systems get wrong is visible to the signature at all. Added and dropped conditions and $\forall/\exists$ choice dominate. Quantifier scope and cardinality, the signature's predicted blind spots, together account for 6.9%. Conjunction/disjunction confusion, a blind spot the reviewer found, accounts for 4.0%.

Table 23. Error types among panel-unfaithful adjudicated candidates (primary error, post-stratified weights; unweighted counts in parentheses).

Error type weighted share (count)
added condition 0.249 (74)
$\forall/\exists$ choice 0.218 (60)
dropped condition 0.144 (47)
implication direction / "only" 0.084 (29)
syntax / unparseable 0.058 (17)
quantifier scope 0.056 (16)
conflation 0.043 (16)
conjunction / disjunction 0.041 (14)
wrong constant 0.039 (9)
negation / polarity 0.032 (10)
argument swap 0.015 (4)
cardinality / numeric 0.014 (1)
wrong split 0.008 (5)

[Figure fig3: generated in the report phase, not in this draft]

Systems, parse rates and label balance

Table 24 gives the per-system picture on the held-out confirmation group. Parse rates range from

0.883 (phi-4) to 0.993 (gpt-oss-120b). Phi-4 answered in LaTeX; a formatting fix to the extractor (not a prompt change) raised its parse rate from 68% to 88%. Final labels follow a fixed precedence:

1. L3 panel majority (609 rows);
2. for ProverQA with panel-approved gold, solver equivalence to the original gold (1,323 rows);
3. otherwise, solver equivalence to the audited gold (3,915 rows);
4. otherwise unknown (453 rows).

Table 24. Held-out confirmation group by system: greedy parse rate [95% CI] and final-label balance.

System	parse rate [95% CI]	faithful	unfaithful	unknown	unfaithful share of labelled
llama-3.1-8b	0.917 [0.894, 0.935]	171	456	73	0.727
qwen-2.5-7b	0.934 [0.914, 0.950]	153	484	63	0.760
mistral-small-3.2-24b	0.957 [0.940, 0.970]	235	418	47	0.640
gpt-4.1-mini	0.984 [0.972, 0.991]	308	364	28	0.542
gemini-2.5-flash	0.969 [0.953, 0.979]	319	341	40	0.517
deepseek-v3.1	0.969 [0.953, 0.979]	254	406	40	0.615
gemma-3-27b	0.964 [0.948, 0.976]	263	395	42	0.600
phi-4	0.883 [0.857, 0.905]	197	405	98	0.673
gpt-oss-120b	0.993 [0.983, 0.997]	317	361	22	0.532
all	0.952 (5,999/6,300)	2,217	3,630	453	0.621

By corpus the unfaithful share of labelled rows is FOLIO 0.748, MALLS 0.662 and ProverQA 0.423. By tercile it is 0.523 (bottom), 0.624 (middle) and 0.677 (top). The sampled group parses at 0.981 for gpt-4.1-mini and 0.831 for llama-3.1-8b.

Table 25 shows how far the solver-only status of candidates moves between the original and the audited gold. On the screen's own Logic-LM candidates, proved equivalences fall from 544 against the original gold to 488 against the audited gold. Among greedy MALLS and FOLIO rows, auditing the gold turned 275 unfaithful labels faithful and 104 faithful labels unfaithful.

Table 25. Solver (L1) status distributions.

Population	equivalent (proved)	non-equivalent	non-equivalent, no bijection	unparseable	no audited gold	capped
greedy vs original gold	1,717	703	3,570	301	n/a	9
greedy vs audited gold	1,973	517	3,356	301	151	2
samples vs audited gold	2,022	545	3,608	661	162	2
screen candidates vs original gold	544	88	338	33	n/a	n/a
screen candidates vs audited gold	488	83	378	31	23	n/a

Contamination and transfer groups

The contamination group was smaller than planned. 203 sentences carried a named entity or constant: FOLIO 187 and MALLS only 16, because MALLS gold is almost constant-free. gpt-4.1-mini paraphrased each of them and renamed its entities, and the renaming was applied to the gold. The panel verified 104: 98 FOLIO and 6 MALLS, against a target of at least 120. Most of the 99 drops were "the renamed gold is not faithful to the paraphrase", which reflects residual gold problems. 124 of the 936 renamed candidates carry a `rename_incomplete` flag. No contamination

measurement has been run yet; the group exists for iteration 2.

The transfer group holds 367 formulas from the user's pilot over 63 EU-AI-Act definitions. 289 of them parse. Sixty-six use $\subseteq$ between variables, which is not FOL, and stay unparseable; the rest fail on tokenization or bracket errors. This group is the only exception-rich material in the run, and it has no labels.

Deviations and cost

The artifact recorded its deviations:

- It found the 2606.02837 release on the Hugging Face Hub (DSAVlab-UNIUD/MALLS_test_subset-CURATED, 100 items; DSAVlab-UNIUD/FOLIO_validation-curated, 275 FOLIO-v2-validation instances). The paper's GitHub repository returned 404.
- A parser bug read constants such as `yr2028` as free variables and universally closed them. It affected 8 golds and 9 candidates. It was fixed after L0; all L1/L2 labels were recomputed and the panel corrections re-derived (0 changed). Re-keying the paraphrase verification cost \$0.49 of repeated calls.
- The ASCII `^` is read as `^`, since LLM outputs use it that way.
- Temperature-0 output is not bit-deterministic across OpenRouter providers, so the provider is recorded per row.
- "Heavily conditioned" in the public corpora means connective-heavy, not exception-heavy. The top tercile is driven by multiple connectives, $\oplus$, $\leftrightarrow$ and quantifier nesting.
- The `cgc2lambda` FOL versions of FOLIO and MALLS were downloaded as possible non-LLM systems but not used: their event semantics has no predicate bijection to the gold.

Total spend was \$9.22 over 18,566 calls, against a \$9.30 hard cap:

Phase | \$

generation | 0.718

panel calibration | 0.877

gold audit (L0 + screen audit) | 3.427

L3 adjudication, design sample | 2.458

L3 adjudication, extra top tercile | 0.878

contamination paraphrases and verification | 0.865

1.7 What iteration 1 established

The signature metric (main hypothesis). On real 2023-era outputs for FOLIO-dev, the monotonicity signature ranks candidates better than the whole-formula judge, round-trip similarity, self-consistency, parse rate and the user's structural metrics. It adds a significant increment over the judge. Three measurements undercut the reason it was proposed:

- it ties an LLM asked the same per-concept questions about the formula;
- most of its signal comes from lexical alignment, and the polarity comparison adds no significant increment once alignment and the judge are controlled;
- it is at chance in the top complexity tercile, where it was predicted to win.

It fails the invariance gate through synonym renaming, the reviewer's objection about alignment made concrete. The text side is the weaker half: the LLM probe misjudges 40% of downward items

on MED and HELP. The zero-LLM rule marker does better on MED and at least as well on real items. The signature's predicted blind spots held exactly (cardinality and conjunction/disjunction at 0.50), and the relativized repair did not work on real text.

The alternates. Cross-system agreement through a latent-class model was the strongest gold-free signal (0.853) and the rule's winner. Its lead is partly built into the labels: under Arm A's labelling it falls to 0.772, and it cannot score a lone output. TVJT is the only text-reading metric whose advantage over the judge grows with complexity (+0.199 in the top tercile). It sees scope and cardinality errors that the signature cannot, and it types errors badly. It failed the invariance gate by a wide margin, because rewrites receive different worlds.

Dead ends recorded in this iteration.

- Instance-consequence NLI with a small DeBERTa model. It was near chance beyond the simplest tercile (0.450 and 0.518) and failed three of four gates. The same pairs judged by gemini recovered +0.079, so the consequence construction carries signal that the small NLI model cannot read.
- Restricting TVJT to non-vacuous worlds. It lowered AUROC by 0.035 overall and 0.114 in the top tercile.
- The relativized signature as a fix for conjunction/disjunction swaps. It gave 0.50 on real text; the formula side works, but the text side does not supply the coordinates.
- The pre-registered probe fallback to gpt-4.1-mini. It piloted at 0.32 downward accuracy and was not adopted.
- The complexity-crossover prediction for the signature. It was not supported on FOLIO. The test is weak here because FOLIO has almost no exception-bearing sentences.
- claude-sonnet-4.5 as a panel member. It failed the calibration gate.
- Gold-error flagging by the signature. Precision at 50 was 0.62 to 0.64 against a base rate of 0.58.

What the dataset changed about the screen. The screen's labels are solver equivalence to gold. The dataset shows that, even after the gold audit, about 44% of solver-non-equivalent candidates are faithful (60% on FOLIO), and solver labels agree with the panel only 68% of the time. The screen AUROCs were therefore measured against a label in which roughly a third of the negatives are not errors. Arm B's 67.6% and Arm A's 79% vocabulary-shape share point the same way. The iteration-1 ranking has not yet been re-checked under the audited screen labels. The pre-registered rule makes that re-check the first step of iteration 2, and the audited ranking governs.

Still open.

1. Confirmation on the held-out 700 sentences $\times$ 9 current systems has not run, so no screen result has been tested outside FOLIO-dev and the 2023 systems.
2. System-level Kendall τ was not measurable with three systems.
3. The contamination check has not run.
4. No public sentence-level gold contains long, exception-heavy sentences; the only such material, the user's EU-AI-Act pilot, has no labels.
5. GenV [8] was not benchmarked.

Spend in iteration 1 was \$5.67 (Arm A) + \$1.55 (Arm B) + \$9.22 (dataset) = \$16.44 across three separately budgeted artifacts.

What we have learned so far

This closing section was rewritten on 2026-09-24, after the round-1 report was restored. The iteration-1 sections above are unchanged. The section now also draws on the round-2 results of the earlier run (run_qY2a2IS-WLIs, iteration 2), which tested the round-1 candidates on the held-out set of Section 1.6. Those results are cited from that run's report (3_invention_loop/iter_2/gen_report_text/gen_report_text/report.md, Sections 2.2 and 2.4). For this document we re-read every one of them from the round-2 artifact output files: the frozen held-out confirmation and the repaired-signature analysis. The script checks/verify_round2.py writes the values to checks/verify_round2_out.json.

Round-2 results carried in from the earlier run

Table 26 lists the round-2 numbers the conclusions rest on, with what each one measures. The terms used below are defined here once.

- The *cheap judge* is the whole-formula LLM judge of Section 1.3 (gemini-2.5-flash, fixed prompt, thinking off). The *frontier judge* is gemini-2.5-pro given the same prompt. The *error-typed judge* is gemini-2.5-flash shown the sentence and the candidate and asked to pick one error type from the panel's taxonomy.
- A score is *zero-LLM* when computing it makes no language-model call.
- *Latent-class agreement* is the one-coin latent-class model of Section 1.4 fitted to the agreement among the nine systems' outputs for the same sentence.
- A *stack* is a logistic regression over several scores. Every stack here is *cross-fitted*: each item is scored by a model fitted on sentence-grouped folds that exclude it. The *frozen base* is the stack fixed before round-2 scoring: the cheap judge, parse rate, round-trip NLI, round-trip cosine and the user's structural metrics. The *judge-only base* is the frozen base without the two round-trip scores.
- *Within-sentence detection* is the probability that an erroneous candidate scores below a panel-faithful candidate of the same sentence from another system. It controls for sentence difficulty, but its samples are small.
- The *repaired signature* is the round-2 version of the monotonicity signature: a global aligner, plus a text side that keeps the zero-LLM rule labels except where the gemini-2.5-flash probe consistently says downward. It is therefore not strictly zero-LLM. Its two zero-LLM components are reported beside it: the alignment-only control and the iteration-1 rule-marker signature.

Table 26. Round-2 results of the earlier run on the held-out set (609 panel-adjudicated candidates from 9 systems, unless stated), re-read from the artifact files. The within-sentence rows cover 36 dropped-condition and 24 implication-direction errors that have a faithful partner; the judges were scored only on the items they covered (21 and 11).

Quantity	Value	[95% CI]	Source (round-2 artifact file)
within-sentence, dropped condition: repaired signature	0.853	[0.739, 0.944]	signature_analysis.json, within-sentence block
same: alignment-only control (zero-LLM)	0.852	[0.735, 0.942]	same
same: polarity component alone	0.469	[0.334, 0.606]	same
same: iteration-1 rule-marker signature (zero-LLM)	0.777	[0.629, 0.896]	same

same: cheap judge (gemini-2.5-flash), $n = 21$ | 0.618 [0.439, 0.769] | same
 within-sentence, implication direction / "only": repaired signature | 0.866 [0.684, 0.992] | same
 same: iteration-1 rule-marker signature (zero-LLM) | 0.870 [0.727, 0.971] | same
 same: polarity component alone | 0.832 [0.647, 0.973] | same
 same: alignment-only control | 0.598 [0.445, 0.756] | same
 same: cheap judge, $n = 11$ | 0.770 [0.461, 1.000] | same
 within-sentence, $\forall/\exists$ choice ($n = 56$): repaired signature; cheap judge; error-typed judge | 0.616; 0.662; 0.918 | same
 wrong-gold flag on 700 golds: cross-fitted judge-only model $\rightarrow$ judge plus repaired signature | 0.808 $\rightarrow$ 0.861, +0.054 [0.029, 0.079] | signature analysis.json, wrong-gold block
 wrong-gold flag, single scores: repaired signature; alignment-only; coverage; judge; error-typed judge | 0.673; 0.690; 0.686; 0.840; 0.859 | same
 latent class (one-coin), AUROC on panel labels | 0.756 [0.701, 0.808] | confirmation confirmation_table.json
 latent class increment over the frozen base [judge, parse, round-trip NLI, round-trip cosine, structural] (0.788) | +0.019 [-0.003, 0.043], not confirmed | same
 latent class + rule-marker signature over the frozen base | +0.030 [0.005, 0.057] (stack 0.818) | confirmation analysis.json, stacking
 latent class over a judge-only base [judge, parse, structural] (0.730) | +0.049 [0.017, 0.085] (stack 0.779) | same
 rule-marker signature over the judge-only base | +0.060 [0.020, 0.101] (stack 0.790) | same
 cheap judge (gemini-2.5-flash), AUROC | 0.762 [0.718, 0.804] | confirmation confirmation_table.json
 frontier judge (gemini-2.5-pro), same prompt | 0.825 [0.788, 0.860]; increment +0.034 [0.017, 0.052], confirmed | same
 repaired signature; rule-marker; alignment-only: AUROC | 0.654; 0.642 [0.589, 0.695]; 0.656 [0.604, 0.705] | signature analysis.json; confirmation table
 cost per judge call, flash vs pro | \$0.0000308 vs \$0.00122 (pilot, 39.7 $\times$); \$0.0000313 vs \$0.00118 (realised, 37.8 $\times$) | confirmation llm_stage_info.json

Conclusions

The monotonicity signature does not replace the judge. On the 2023-era screen it ranked candidates above the cheap judge (0.711 to 0.718 against 0.644, Table 4). On held-out outputs from current systems the order reversed: the cheap judge scored 0.762 and the signature variants 0.642 to 0.656. The signature did no better than its own alignment-only and coverage controls. In neither round did its polarity comparison add significant signal once lexical alignment was controlled.

Its zero-LLM checks target specific error types, and they do it through different components. Within a sentence, the repaired signature detects dropped conditions at 0.853 and implication-direction errors at 0.866. The cheap judge scores 0.618 and 0.770, on only 21 and 11 items with wide intervals.

- **Dropped conditions: the aligner, not polarity.** The alignment-only control matches the signature (0.852), and polarity alone scores 0.469. A dropped condition leaves a sentence concept with no predicate, so this detection is concept coverage, not monotonicity.
- **Implication direction: polarity.** Polarity alone scores 0.832, and the zero-LLM iteration-1 rule marker scores 0.870.

On the $\forall/\exists$ errors that current systems make most often, the signature is weak: 0.616 within

sentence, against 0.918 for an error-typed judge.

For wrong-gold flagging the signature complements a judge. Adding the repaired signature to a cross-fitted judge-only model raised gold-flag AUROC from 0.808 to 0.861 (+0.054 [0.029, 0.079]). That puts it level with the error-typed judge alone (0.859). By itself the signature flags wrong gold at 0.673, no better than its alignment-only (0.690) and coverage (0.686) controls.

Latent-class agreement is the strongest signal that does not read the text, and it complements a cheap judge in combination. It scored 0.853 on the screen and 0.756 on the held-out set. Alone it did not add significantly to the frozen base (+0.019 [−0.003, 0.043]), so round 2 did not confirm it. Together with the rule-marker signature it added +0.030 [0.005, 0.057], reaching 0.818. It also adds +0.049 to a judge-only base, reaching 0.779. The frozen base already contains round-trip features, which cost more than the judge itself.

The frontier judge is the only confirmed metric, at about 40 times the cheap judge's cost per call. gemini-2.5-pro with the same prompt scored 0.825 [0.788, 0.860] and added +0.034 [0.017, 0.052] to the frozen base. Per call it cost $37.8\times$ (realised) to $39.7\times$ (pilot) as much as gemini-2.5-flash.

The round-1 findings still stand:

- The signature's predicted blind spots came true: cardinality and conjunction/disjunction detection sit at 0.50 on mutants.
- Solver-exact formula semantics raised no false alarms on logical rewrites; synonym renaming breaks the lexical alignment.
- Instance-consequence NLI with a small model is null.
- Public gold is unfaithful on 44% to 62% of FOLIO and MALLS items, and solver equivalence to shipped gold marks about 44% of faithful non-equivalent candidates wrong. That bounds every screen number above.

Taken together, the evidence supports a narrower role than the hypothesis proposed. Zero-LLM coverage and polarity checks, plus latent-class agreement, are candidate complements to a cheap judge on dropped conditions, implication direction and gold errors. They have not been shown to reach a frontier judge. The one combination that came near it also contained round-trip features.

Next steps: one pre-registered round-2 test

This test is registered here, before any of its data exist. It asks whether the complement described above holds on sentences that the earlier run's round 2 did not score.

Claim under test. A stack of the cheap judge plus zero-LLM checks meets two conditions:

1. It beats the cheap judge on dropped-condition and implication-direction errors.
2. It comes within 0.02 AUROC of gemini-2.5-pro, at no more than 1/40 of gemini-2.5-pro's cost per scored candidate.

We call this the complement claim.

Population. New sentences only. The following are excluded:

- the 700 held-out sentences of Section 1.6;
- the 360 screen sentences;

- the 104 contamination sentences;
- any sentence used for MED/HELP development or any round-2 tuning.

The sentences are drawn from the remaining eligible pool of the Section 1.6 recipe: MALLS-v0.1-test, FOLIO-v2-train stories disjoint from both validation sets, and ProverQA-dev medium/hard. The same complexity stratification applies, with the top tercile oversampled. Candidates come from the same nine systems with the frozen generation prompt.

Labels. Blinded three-model panel adjudication under the round-1 protocol and calibration gate, family-disjoint from the gemini judges. The panel records a primary error type, and estimates use post-stratified weights. Sampling continues until each target error class has at least 50 unfaithful items with a panel-faithful partner in the same sentence. A class that falls short within the labelling budget is reported as underpowered and counts as not passed.

Frozen scores.

- Cheap judge: gemini-2.5-flash, fixed prompt, thinking off.
- Frontier judge: gemini-2.5-pro, same prompt.
- Checks, all zero-LLM: the one-coin latent-class posterior over the nine systems; the iteration-1 rule-marker signature (polarity); the alignment-only control (concept coverage).
- Secondary, reported but not deciding: the repaired signature, which consults gemini-2.5-flash.

The stack is a logistic regression on [cheap judge, parse rate, checks], cross-fitted over sentence-grouped folds. Its features, folds and regularisation are frozen now. It contains no round-trip features.

Tests. Four tests, each a sentence-clustered paired bootstrap with 2,000 draws and one-sided $\alpha = 0.05$. The first two together are the error-class tests.

- **Dropped-condition test.** The stack's within-sentence detection of dropped-condition errors exceeds both the cheap judge's raw score and a cross-fitted judge-only model. The lower bound of each paired difference must be above 0.
- **Implication-direction test.** The same comparison for implication-direction and "only" errors.
- **Frontier-gap test.** On all panel items, the lower bound of the stack's AUROC minus gemini-2.5-pro's AUROC is above -0.02 .
- **Cost test.** The stack's realised API cost per scored candidate is at most $1/40$ of gemini-2.5-pro's realised cost per call.

Latent class needs other systems' outputs. The cost test is decided on the accounting in which those outputs already exist, for example when all nine systems' candidates are scored together. A second accounting is also reported, in which each extra generation is charged at the Section 1.6 recipe's cost: \$0.7177 for 13,300 outputs, about \$0.000054 each. Scoring one system's output alone then needs eight extra generations, about \$0.00043. That is roughly a third of gemini-2.5-pro's \$0.00118 per call, so the cost test fails under that accounting.

Decision.

- All four tests pass: the complement claim is supported.

- Both error-class tests pass and the frontier-gap test fails: the checks are reported as error-type detectors that do not close the gap to the frontier judge.
- Either error-class test fails: the within-sentence advantage in Table 26 is reported as not replicated.

Prior evidence, stated before the test. On the earlier run's 609 panel items, round-2 stacks without round-trip features stayed 0.035 to 0.046 below gemini-2.5-pro:

- judge-only base + latent class: 0.779;
- judge-only base + rule-marker signature: 0.790.

Only the stack that also contained flash round-trip features came within 0.02 (0.818 against 0.825). Round-trip adds a flash back-translation call of about \$0.000118 per item, so that stack costs about 1/8 of gemini-2.5-pro, not 1/40. The round-2 evidence therefore predicts that the frontier-gap test fails at the 1/40 cost point. The error-class comparisons rest on 21 and 11 judge-scored items, too few to predict either way. The test is registered to settle the claim, not because the prior evidence favours it.

Still open from Section 1.7:

- GenV [8] has not been benchmarked;
- no public gold contains long, exception-heavy sentences;
- the user's EU-AI-Act pilot has no labels.

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